

Jonas Michael Fritsch



Personal

Jonas Michael Fritsch
Born in Schwaebisch Hall
on twenty five zero five two
thousand two

Specialization

Mechanical Engineering
• Robotics • Control
Engineering • Kinematics

Hobbies

Skiing • Squash • Bartending

 Jonas Fritsch

Experience

03/2025-09/2025

Chair of Product Development and Lightweight Design Semester Project · Garching



Development of a flexible tactile skin for real time sensing of normal forces shear forces and position:

- Concept and development of a tactile skin for capturing normal forces shear forces and position
- Integration of different sensor concepts investigation of accuracy and scalability
- Goal improvement of gripping processes in real time without damage to the objects

04/2024-present

Blechverarbeitung Michael Fritsch GmbH



Working Student · Hesselental

Work in machine programming and support in manufacturing

09/2023-04/2024

Gerhard Schubert GmbH



Internship · Crailsheim

Internship in the department for new development of standard modules:

- Execution of FEM and motion simulations using SolidWorks
- Design of slide bearings bearings and hinges for assemblies
- Execution of friction tests and evaluation
- Revision of a conveyor belt for a vertical magazine
- Development of an interface construction for folding box and stamping tool
- Optimization of an assembly unit to sheet metal construction

10/2023-04/2024

Bartenders



Barkeeper · Crailsheim

Bartending and service work in the evening gastronomy

03/2023-09/2024

wbk Institute of Production Science KIT



Bachelor Thesis · Karlsruhe

Systematic analysis and optimization of the workflow for robot programming using external control in the context of additive repair:

- Development of an external control of a Kuka-Kr 6 through Ethernet TCP Master Client with Kappa-K interface
- Programming of the external robot control in C # and implementation of a Matlab interface for command verification
- Development of a new six point calibration method for precise module calibration
- Comparison with the KUKA three point method and validation through tests with higher measurement accuracy

2017-09/2023

Blechverarbeitung Michael Fritsch GmbH



Working Student plus Internships plus Vacation Jobs · Hesselental

Working student internships and vacation jobs:

- CNC machine programming for flat bed laser bending machine and robot cells
- Support in manufacturing and assembly design
- Extensive practical experience laser punching welding TIG/Mag bending grinding powder coating

Degrees

- 2020 **High School Graduation**
Abitur · Gymnasium bei
Sankt Michael 
- 2024 **Mechanical Engineering**
B.Sc. · Karlsruhe Institute of
Technology 

Languages

German	C2	native
English	C1	fluent
French	B1	basic knowledge

Academic Path

- 09/2025–present **Robotics** 
Exchange Semester · Barcelona 
Studies at the School of Industrial Engineering of Barcelona E T S E I B
U P C Universitat Politècnica de Catalunya
- 2024–present **Mechatronics Robotics and Biomechanical Engineering** 
Master · Munich 
Expected completion in two thousand twenty six at the Technical Uni-
versity of Munich
- 2020–2024 **Mechanical Engineering** 
Bachelor · Karlsruhe 
Bachelor of Science at Karlsruhe Institute of Technology
Final grade two comma two
- 2012–2020 **General Higher Education Entrance Qualification** 
Student · Schwäbisch Hall 
High school graduation at Gymnasium bei Sankt Michael
Final grade two comma zero

Software Skills

- CAD **PTC Creo**
Advanced knowledge in part assembly and drawing creation expe-
rience with top down design and skeleton technique
SolidWorks
FEM and motion simulation mechanical design and optimization of
assemblies
Fusion 360
Three D modeling sheet metal design and visualization
- Code **C#**
Development of a TCP Master Client communication for external
robot control KUKA Kappa-K
MATLAB
Creation of an interface for command verification data analysis and
visualization
C++
Basic knowledge in automation and control engineering

Certificates

- 2022 **Advanced Certificate CAD PDM KIT**
Top down design P D M system Windchill visualization and rendering
- 2021 **Basic Certificate CAD KIT**
Part assembly and drawing creation with Creo
- 2021 **Sheet Metal Design Workshop Trumpf+KIT**
Development of a sheet metal oriented product in a team practice oriented
project